Look Again: Self-Correcting Tool Use for Panoramic Dental Radiograph Diagnosis

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Abstract

Vision-language models (VLMs) fail structurally on panoramic dental radiographs: the best frontier models reach only 36–46% exact-match accuracy against expert labels on the DENTEX benchmark while reporting 0.85–0.95 confidence on hallucinated tooth positions (ECE up to 0.68), and a dense 11B generalist falls below the statistical majority class — because one forward pass cannot zoom into sub-patch pathology, resolve the modality's left/right display convention, or verify an anatomical hypothesis before committing to it. We present **VLM-DENTAL**, an agentic VLM built on a different contract between the policy and its tools: *grounding is a hypothesis to interrogate, not an answer to trust.* (1) The policy can correct its learned grounding tool via `nudge_crop`, a data-only tool taking box-relative fractional shifts and rescales that a VLM can reason about linguistically. (2) A *tiered perturbation* mechanism makes this skill learnable from synthetic demonstrations: the demonstrator experiences deliberately imperfect tool output (45% clean / 25% small / 30% large offsets, decoupled from the detector's real error rate so demonstrations never go stale, and never announced verbally so the offset magnitude alone carries the signal). (3) The eight-tool interface exposes continuous effect parameters (denoise strength, window center/width, contrast factor, crop context), so tool calls carry *how much*, not just *which*. (4) GRPO reinforcement learning uses a ground-truth-anchored composite reward with multi-finding set matching, whose efficiency term exempts locate-and-correct calls so the reward never prices out the behavior it teaches. A corpus-level analysis of 3,694 verified, decontaminated synthetic traces shows a clean judgment gradient — corrective nudges on 38.4% of large tool errors, 13.6% of ambiguous ones, 1.1% of clean outputs (ground-truth-anchored pathology locates), with recoveries up to IoU 0.07 → 0.94 — genuine visually-driven self-correction rather than scripted steps. Evaluation follows DENTEX-native, leaderboard-comparable set-matched metrics with paired bootstrap intervals and a frozen, leak-free protocol; end-to-end training results are reported as runs complete. We position the work relative to concurrent dental agentic systems and release the corpus, audit tooling, and all figure/table sources for regenerability.

1 Introduction

Dental panoramic radiographs (orthopantomograms, OPGs) are the workhorse imaging modality of general dentistry, and annotating them — *which tooth, and what is wrong with it*, in standard FDI notation — is the task that ultimately feeds treatment planning. It is also a task on which general-purpose vision-language models (VLMs) visibly fail. Frontier multimodal models evaluate at best 36–46% exact-match accuracy against expert labels on the DENTEX benchmark and carry expected calibration errors of up to 0.68, reporting 0.85–0.95 confidence while hallucinating tooth

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positions (Section 4.2); a dense 11B generalist collapses below the statistical majority-class floor. The failure is not primarily a knowledge gap. An OPG is a single very wide image in which an early interproximal carious lesion occupies fewer than 15×15 native pixels — reduced to a sub-patch blur by the vision encoder's resolution budget — and in which left/right inversion conventions routinely flip quadrant labels. What the task rewards is exactly what a radiologist does and a single-pass VLM cannot: *look again* — zoom into a suspect region, adjust contrast, compare against the contralateral jaw half, and verify an anatomical hypothesis before committing to it.

A growing body of work equips VLMs with callable tools to close this gap (Wu et al., 2026; Li et al., 2024; Lu et al., 2025), and two concurrent systems do so for dentistry specifically (Fan et al., 2026; Hao et al., 2026). But in all of them the tools the model calls are *trusted* operations: a zoom returns a crop, a mirror returns a composite, and the model has no way to tell the tool that its output is wrong. This is a structural mismatch with the medical setting, where the most informative component the system can call — a learned grounding model that says "tooth 36 is here" — is precisely the component whose errors the agent must anticipate rather than inherit. A grounding box that is off by half a tooth-width does not merely lose accuracy; it silently poisons every downstream observation, because the crop looks authoritative.

This paper develops and evaluates VLM-DENTAL, an agentic VLM for OPG diagnosis built around a different contract between the policy and its tools: *grounding is a hypothesis, not an answer*. Fig. 3 gives the complete system architecture, and Fig. 10 contrasts the two trace arms on one image. Our contributions are:

1. **A self-correcting grounding tool (`nudge_crop`, Section 3.4).** The policy can shift and rescale a box it was *already given* — by the grounding model or by a previous correction — using box-relative fractional arguments (`dx_frac`, `dy_frac`, `scale`) that a VLM can reason about linguistically ("shift right about half a tooth-width"), without re-running detection. To our knowledge neither concurrent dental agentic system (Fan et al., 2026; Hao et al., 2026) offers the policy any mechanism to correct a specialist tool's own uncertainty.

2. **A tiered perturbation mechanism that makes self-correction learnable (Section 3.5).** Demonstrating correction behavior requires the demonstrator to *experience* tool error. During trace generation we return the expert-verified box for annotated teeth (guaranteeing task solvability) but deliberately perturb the box *shown to the demonstrator* — 45% clean, 25% small offsets (12–28% of box size), 30% large offsets (45–75%) — with two design commitments: the rates are fixed constants decoupled from the detector's real error rate, so demonstrations never go stale as the detector improves; and errors are never announced verbally, so the offset magnitude alone must carry the signal and the learned behavior survives to reinforcement learning and inference time, where no such cue exists. A corpus-level analysis (Section 4.4) shows the mechanism elicits exactly the intended behavior gradient in 3,694 generated demonstrations: corrective use of `nudge_crop` on 38.4% of large offsets, 13.6% of small ones (a genuine judgment call), and 1.1% of clean boxes, on ground-truth-anchored pathology locates.

3. **A tool interface with continuous parameters (Section 3.3).** Tools expose *how much* effect to apply, not just which tool to call: `denoise(strength` $\in [0, 1]$`)`, `window_level` with exact `center`/`width` values as an override on top of presets, a multiplicative `enhance_contrast` factor, and a tunable `zoom_crop` context margin. Tool calls therefore carry real information beyond tool identity, and the policy can match processing intensity to image characteristics.

4. **A ground-truth-anchored, correction-friendly reward (Section 3.8).** GRPO training (DeepSeek-AI, 2025) uses a composite reward whose dominant term is a hard, non-LLM-judge match against expert labels with multi-finding greedy one-to-one matching (F1-style harmonic

mean of recall and precision), and whose efficiency term scales a reference budget with case complexity and *exempts* `locate_tooth` and `nudge_crop` from per-call cost — so the reward never prices out the verification behavior it is trying to teach, a failure mode we identified and removed by design.

5. **A fully synthetic, clinician-free data pipeline and its audit trail (Section 3.6).** 3,694 verified chain-of-thought tool-use traces over DENTEX and Tufts (1,847 with tools, 1,847 single-turn without, exact 1:1 parity), generated by frontier models under real tool execution in a LangGraph loop, verified cross-family, decontaminated of directive leakage (105 traces regenerated; 0 leaks in the canonical corpus), and released as a reusable resource. The verifier's rejection taxonomy (66 false-positive findings on healthy scans, 15 ground-truth contradictions, 11 uncorrected spatial drifts) doubles as an audit of what the perturbation mechanism does and does not elicit.

We evaluate on DENTEX with its native structured metrics — pooled, set-matched, finding-level precision/recall/F1 with paired bootstrap intervals — against zero-shot frontier VLMs, the same backbone with and without tools, and a supervised specialist detector. The training-side analyses in this draft are complete; the two-stage training runs (SFT, then GRPO) are in progress, and their result cells are marked **[SFT/GRPO results pending first hardware run]** rather than extrapolated. The zero-shot baseline table, the grounding-tool validation, and the corpus-level mechanism analysis are final.

2 Related Work

Supervised dental AI. The dominant paradigm in dental radiograph analysis is narrow, single-task supervised learning: CNN and Faster R-CNN cephalometric landmark detectors, U-Net-style segmentation, FDI tooth-numbering classifiers, and YOLO-family or Faster R-CNN detectors for caries, periapical lesions, and impactions (Hamamci et al., 2024). The DENTEX benchmark consolidated this into a hierarchical quadrant–enumeration–diagnosis task with expert-verified labels (Hamamci et al., 2024) and remains the most structured public benchmark for our problem. Across this literature the model is a single forward pass trained end-to-end on labels; none of it plans, calls tools, or can request a closer look at an ambiguous region. A systematic review of public dental imaging datasets (Uribe et al., 2024) screened 131,028 records and found only 16 datasets meeting a minimal bar — the data scarcity that motivates our synthetic-trace approach.

Reinforcement-learned reasoning in medical VLMs. Following the GRPO recipe (DeepSeek-AI, 2025), Hu et al. (2025) fine-tuned a 2B VLM with GRPO on a few hundred examples and reported large gains over zero-shot across MRI, CT, and X-ray VQA; Med-R1 extended this across eight modalities and found that lengthening the chain of thought does not reliably improve accuracy — reward alignment matters more than reasoning length (Lai et al., 2026). HuatuoGPT-o1 (Chen et al., 2024) and OpenMedReason (Baghbanzadeh et al., 2026) build reasoning supervision by using a frontier model purely as a *verifier* of evidence-grounded traces. None of this work calls tools: the reasoning is textual, conditioned on one static image.

Agentic tool use in VLMs. MMedAgent trains an LLM to select among grounding, segmentation, and report tools (Li et al., 2024); OctoTools generalizes the orchestration pattern (Lu et al., 2025). Closest in spirit is VTool-R1 (Wu et al., 2026), which trains a VLM with GRPO to call Python-based visual-editing tools mid-reasoning, improving chart/table QA from 64.7% to 71.7%. Its tools are

generic image-editing primitives, its reward is a single outcome-accuracy term, and its domain is not medical. Two concurrent dental systems must be addressed directly. **OralGPT-Plus** (Fan et al., 2026) pairs trained CoT, two callable image tools (zoom-in, contralateral mirror), and GRPO-style RL with a composite reward — the closest concurrent work, but it evaluates on its own benchmark with an LLM-judge-scored reward (whose documented reward-hacking failure mode motivates our ground-truth-anchored accuracy term, Section 3.8) and does not cleanly separate tool-use from RL effects. **OralAgent** (Hao et al., 2026) is a broader prompted ReAct orchestrator around an off-the-shelf model, not an RL-trained tool-use policy. Neither system gives the policy a way to correct a tool's output — the mechanism this paper is built around. We therefore position this work as *among the first* RL-fine-tuned tool-augmented agentic VLMs for dental imaging, and specifically the first with DENTEX-native, leaderboard-comparable evaluation, a clean two-factor ablation design (tools vs. RL), and a self-correcting grounding loop.

Synthetic reasoning traces without clinicians. Frontier LLMs can generate and verify reasoning-trace training data of sufficient quality for downstream fine-tuning (Chen et al., 2024; Baghbanzadeh et al., 2026), replacing prospective expert annotation with expert-*verified benchmark labels* plus cross-model automated checks. We extend this pattern to *multi-turn tool-use* traces: the demonstrator operates a real tool suite under real tool feedback (including deliberately imperfect tool output, Section 3.5), and an independent cross-family verifier audits every trace against expert labels before it enters the training set.

3 VLM-DENTAL: System and Method

3.1 Problem setup and agent loop

An OPG I is annotated as a *set* of findings $F = \{f_1, \ldots, f_m\}$ (possibly empty for a healthy image), where each finding $f = (\text{quadrant}, \text{tooth position}, \text{diagnosis})$ uses FDI two-digit notation over DENTEX's four diagnosis classes (caries, deep caries, periapical lesion, impacted tooth). The agent receives the image and a registry of tools, and produces a sequence of turns; each turn contains free-form clinical reasoning and zero or more structured tool calls, and the loop ends with a structured final answer reporting *every* finding the agent commits to. The same agent loop is used for synthetic data generation, RL rollouts, and evaluation, and tool calls execute for real against the source image at every stage (Fig. 1); a single shared dispatcher defines which tools consume the image and which image they operate on, so the training and inference environments cannot drift apart. Tool calls are capped at 50 per episode, counted by calls rather than turns, and parsed by a bracket-counting JSON extractor that repairs truncated or markdown-wrapped outputs.

3.2 System architecture

Fig. 3 shows the complete system. Five component groups surround one agent core — a Qwen3.5-9B vision-language policy running the loop of Section 3.1: a data layer (Fig. 2: registry-driven loaders for the two in-use sources, DENTEX and Tufts; a candidate pool of eight further public panoramic datasets, promoted only after their annotation semantics are verified; and surgical per-image Hugging Face caching), the trace synthesis stack of Section 3.6, the two training stages of Section 3.8, and an evaluation layer (Section 4.1) whose probes feed model selection on a fixed dev split.

Three single-source-of-truth decisions hold the system together. First, one shared dispatcher defines, for every tool, whether and *which* image it consumes — always the native source image — so the trace-generation loop, the GRPO rollout loop, and the evaluation runner cannot silently diverge;

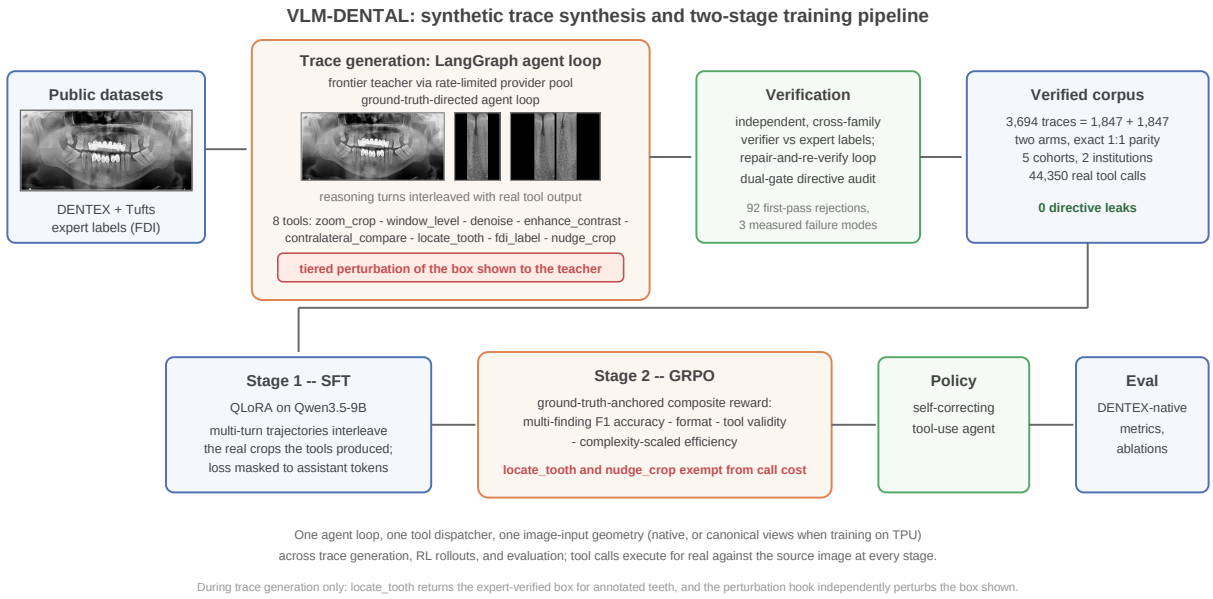


Figure 1: The VLM-DENTAL pipeline. Traces are synthesized by a frontier teacher operating a real tool suite in a LangGraph loop (with the tiered perturbation hook of Section 3.5), verified cross-family against expert labels, and used for SFT followed by GRPO with a ground-truth-anchored composite reward. Tool calls execute for real against the source image at every stage.

the dispatcher exists precisely because the two loops originally duplicated this logic and the rollout side drifted onto a compounding image state, a real bug that silently invalidated rollouts. Second, the image-input geometry lives in one module and is an explicit flag, not an implicit behavior (Fig. 4). The two candidate execution targets make different demands: TPU training requires aspect-preserving canonical views with static vision-slot padding so XLA compiles a single graph and never recompiles, while GPU runs so far feed original images with dynamic shapes; which target the project settles on — a GPU (RTX 4090 / RTX Pro 6000 / A100) or a TPU v5e-8 cluster — is deliberately deferred to the first GPU run. Whichever is chosen applies identically across SFT, RL, and evaluation, and in both modes every tool operates on native-resolution pixels, so coordinates the model was trained to emit remain valid at inference. Third, the conversion from DENTEX's zero-indexed annotation rows to FDI notation has exactly one implementation, called by the reward, the verifier, and the metrics alike — an indexing bug cannot recur independently in several subsystems.

3.3 A tool suite with continuous parameters

The registry exposes eight tools (Table 1). The deliberate design choice throughout is that tools expose *how much* effect to apply as a continuous or overridable parameter, not just a fixed on/off call: denoising takes a continuous `strength`, windowing takes exact `center`/`width` values as an override on top of tissue presets, contrast takes a multiplicative `factor`, and cropping takes a tunable context margin. A policy calling `denoise(strength=0.3)` versus `denoise(strength=0.9)` is expressing a patient-specific judgment about how grainy a crop is — information a fixed three-level menu cannot carry — and the resulting tool-call stream is correspondingly more informative to train on and to audit.

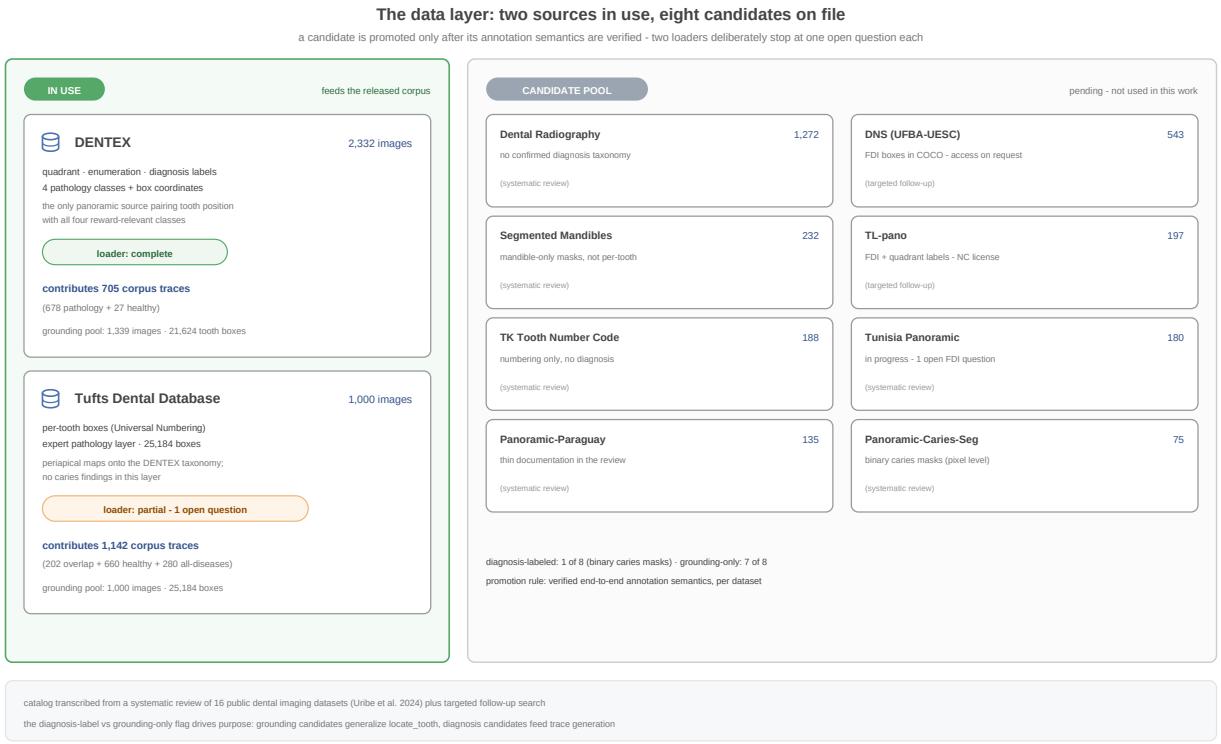


Figure 2: The data layer. Two sources are in use — DENTEX (the only panoramic dataset pairing tooth position with all four reward-relevant pathology classes) and the Tufts Dental Database (per-tooth boxes plus an expert pathology layer whose periapical class maps onto the DENTEX taxonomy). Eight further public panoramic datasets are catalogued as candidates and deliberately *not* used: each would be promoted only after its annotation semantics are verified end-to-end, and two loaders currently stop at exactly one open question rather than guess. The diagnosis-label vs grounding-only flag drives purpose: grounding candidates generalize `locate_tooth`, diagnosis candidates would feed trace generation.

3.4 Self-correcting grounding: `nudge_crop`

The grounding tool is a YOLOv8m detector fine-tuned on 2,339 DENTEX+Tufts images with 46,808 tooth boxes, trained with 5-fold cross-validation (one class per FDI quadrant–position pair) (Fig. 5; full per-fold numbers in Section C). It is strong — held-out *target* mAP50 of 0.93–0.96 with precision up to 0.986 on annotated teeth — but it is a 32-class detector operating under per-fold variance, and its errors are exactly the kind that poison downstream reasoning: a box shifted by half a tooth-width produces a crop that still *looks* like a plausible anatomical region while showing the wrong tooth.

`nudge_crop` turns that failure mode into an interaction. It takes a box the policy was *already given* (by `locate_tooth` or a previous `nudge_crop` — never asserted from nowhere) and returns adjusted coordinates: a shift (`dx_frac`, `dy_frac`) expressed as fractions of the box's own width and height, and a rescale `scale` around the new center, clamped to image bounds. The tool is deliberately *data-only*: it computes coordinates, and the policy must pair it with `zoom_crop` to actually see the corrected region. Two consequences follow. First, the fractional, box-relative parameterization is what a VLM can reason about reliably — "shift left by about half the box" — rather than raw pixel coordinates at an arbitrary zoom level. Second, because each correction must be followed by a verification crop, the loop enforces an inspect–correct–re-inspect cycle rather than a single blind adjustment. This is the mechanism neither concurrent dental agentic system provides:

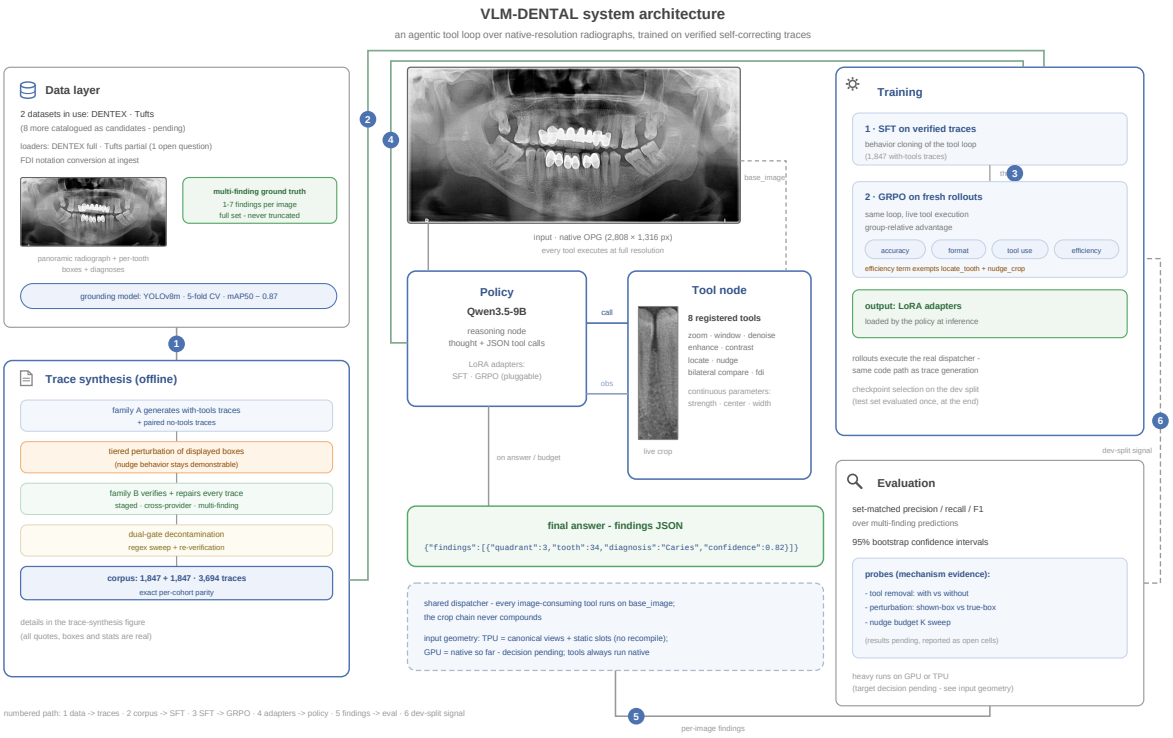


Figure 3: The VLM-DENTAL architecture. One agent loop, one shared tool dispatcher, and one canonical vision geometry are shared by trace generation, RL rollouts, and evaluation; the trace synthesis stack produces the verified corpus consumed by SFT and GRPO; evaluation probes select on a dev split only, with the test split run once at the end. Numbered badges trace the corpus-to-policy life cycle (legend in the figure); radiograph and crop pixels are real assets from the corpus.

Table 1: The VLM-DENTAL tool suite. Continuous or overridable parameters are shown in *italics*; data-only tools return coordinates rather than images.

Tool	Key parameters	Purpose
`zoom_crop`	bbox, *padding_frac*	High-resolution crop of a region with a tunable context margin
`window_level`	preset; *center*, *width*	Radiological intensity windowing (bone/enamel/soft-tissue), exactly overridable
`denoise`	method, *strength* $\in [0, 1]$	Edge-preserving bilateral or median filtering matched to crop noise
`enhance_contrast`	*factor*	Multiplicative contrast adjustment for subtle lesions
`contralateral_compare`	bbox, quadrant	Side-by-side mirror crop, constrained to the correct jaw half, for bilateral symmetry assessment
`locate_tooth`	FDI tooth	YOLOv8m grounding (32 classes, one per FDI position); returns a box
`fdi_label`	quadrant, position	FDI notation conversion and validation
`nudge_crop`	bbox, *dx_frac*, *dy_frac*, *scale*	Correct a box the agent was already given, without re-running detection

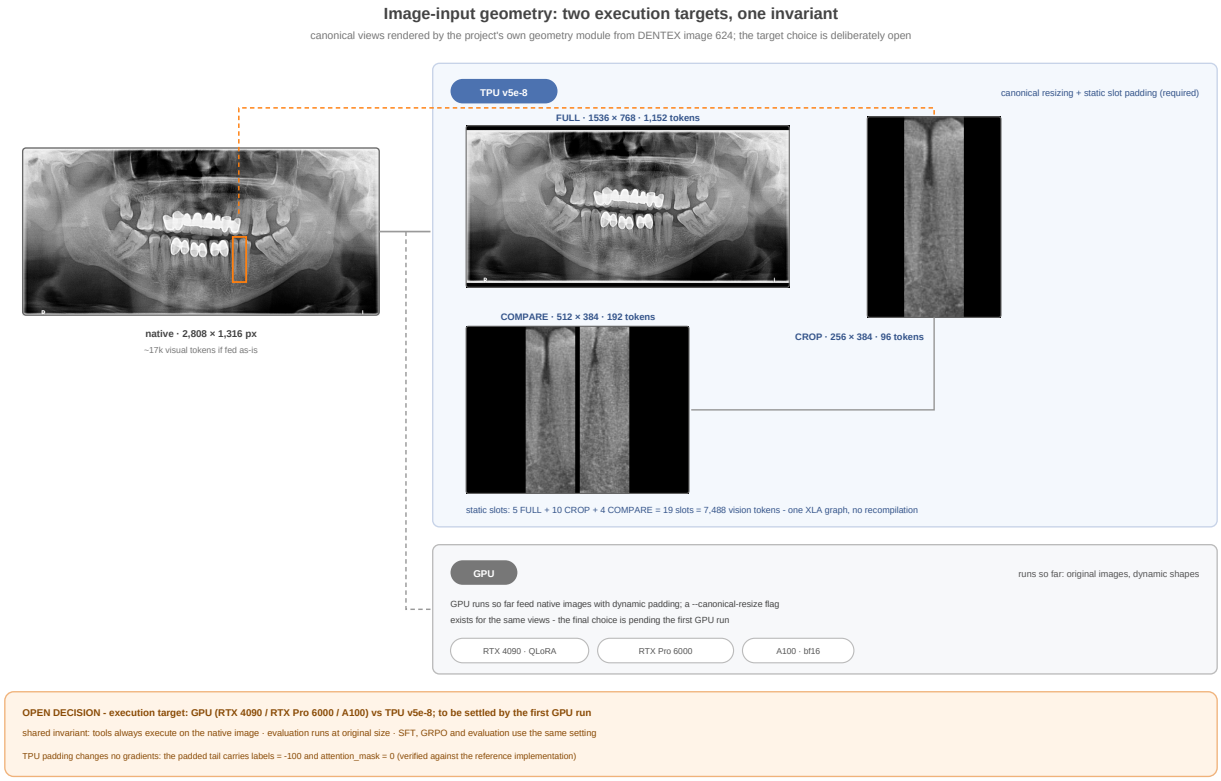


Figure 4: Image-input geometry: two execution targets, one invariant. **TPU:** the policy sees aspect-preserving canonical views in three fixed families (renders produced by the geometry module itself, and the `contralateral_compare` tool, from a DENTEX image), padded to a static 19-slot budget of 7,488 vision tokens so XLA compiles one graph; the padded tail carries labels $= -100$ and attention_mask $= 0$, so padding changes no gradient. **GPU:** runs so far feed original images with dynamic shapes. The choice between the targets is explicitly open pending the first GPU run; in both, tools always execute on the native image and evaluation runs at original size.

their tools are fixed operations whose outputs the policy can only accept.

3.5 Tiered perturbation: making self-correction learnable

A policy can only learn to interrogate its tools if its teachers did. The core difficulty is that an idealized demonstrator — one whose grounding tool always returns the right box — demonstrates *trusting* the tool, and SFT on such traces teaches exactly the wrong reflex. At the same time, blind exploration by the demonstrator would collapse the yield of usable traces.

Our resolution (Fig. 6) separates what the grounding tool *computes* from what the demonstrator *sees*. During trace generation, `locate_tooth` operates in a ground-truth-grounding mode: for an annotated tooth it returns the expert-verified box (guaranteeing the demonstration remains solvable), while for exploratory calls on unannotated teeth it falls back to real detector inference. Then, independently of how the box was obtained, a *perturbation roll* decides what is displayed:

- **none** (45%): the box is shown as computed — a clean accept-and-proceed demonstration;
- **small** (25%): a synthetic offset of 12–28% of the box's own size per axis, random sign — a genuine judgment call that the zoom margin often absorbs, and that the prompt deliberately does *not* resolve either way;

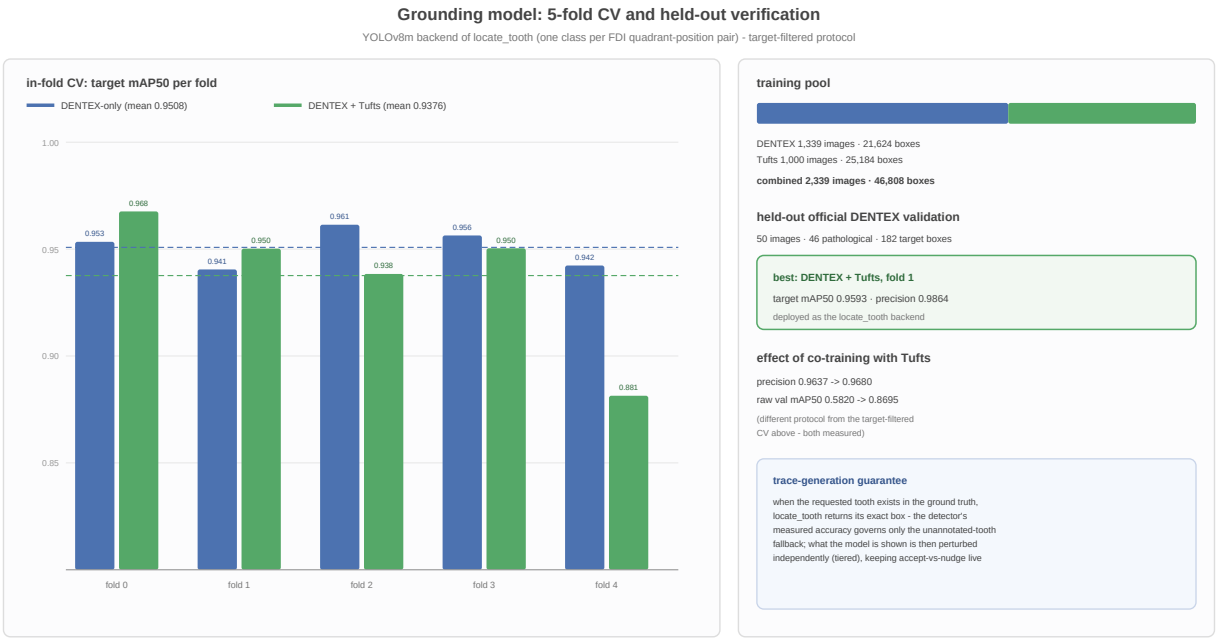


Figure 5: The `locate_tooth` backend: YOLOv8m trained with 5-fold cross-validation on the combined DENTEX+Tufts pool (2,339 images, 46,808 tooth boxes). Left: in-fold target mAP50 per fold for the DENTEX-only and multi-dataset models (means 0.9508 / 0.9376). Right: pool composition, the held-out official DENTEX validation (best target mAP50 0.9593 at precision 0.9864 — the deployed checkpoint), and the trace-generation guarantee: when a requested tooth exists in the ground truth, `locate_tooth` returns its exact box, and only the *displayed* copy is perturbed. Exact per-fold numbers in Section C.

- **big** (30%): an offset of 45–75% per axis — sized to be visually self-evident.

Offsets are applied with `nudge_crop`'s own shift-and-clamp math, so every perturbed box is one the correction tool could plausibly have produced or fixed; the unperturbed box is logged internally for auditing but never shown. Two design commitments matter more than the numbers:

Fixed rates, decoupled from detector accuracy. The perturbation probabilities are configured constants, not a reflection of the grounding model's measured error rate. This is a deliberate decoupling: if "how often the model must verify" tracked the detector's live precision, then every improvement to the detector would silently reduce the frequency of demonstrated correction, and traces generated under the old detector would teach a stale behavior. With fixed rates, today's corpus remains valid as the grounding tool improves — a property this project exercised for real, as the detector's validated mAP50 improved from 0.58 (DENTEX-only) to 0.87–0.96 (multi-dataset) while the trace corpus stayed in use unchanged.

No verbal cue; magnitude carries the signal. Nothing in the trace-generation prompt marks a perturbed box as wrong. A verbal cue ("this box needs correction") would exist only during data generation and not at RL or inference time, so teaching the model to respond to it would be a shortcut that fails exactly when it matters. The big tier is therefore sized to be self-evident from the crop alone, and the small tier is left genuinely ambiguous — the demonstrator has to look and judge, which is precisely the skill the deployed policy must exercise with no ground truth available at all.

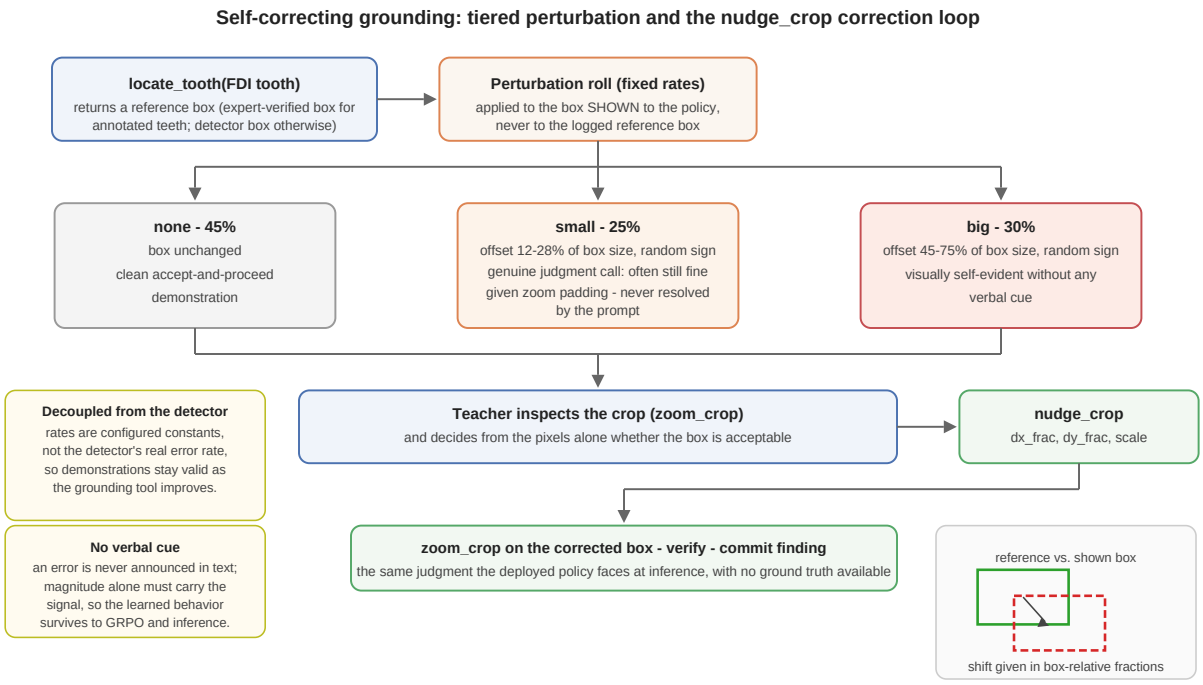


Figure 6: The tiered perturbation mechanism and the correction loop it trains. The perturbation applies to the box shown to the policy only; offsets reuse `nudge_crop`'s own math; and no verbal cue marks an error, so the offset magnitude alone must carry the signal.

3.6 Trace generation, verification, and decontamination

Traces are generated by a frontier teacher (Gemini 3.5 Flash Lite or MiniMax M3, routed through a rate-limited provider pool) that is directed toward the expert labels for the image but writes its own reasoning and tool calls turn by turn under real tool feedback. Generation is ground-truth-directed by design: blind exploration by an untuned model would tank the yield of correctly-labeled demonstrations. Every trace then passes an independent, cross-family *verifier* conditioned on the image and the expert labels, which rejects reasoning not supported by what the model was actually shown. Verification is staged rather than monolithic (Fig. 7): a strict pass checks diagnostic correctness, FDI validity, and whether every claim is supported by what the model was actually shown — treating valid `nudge_crop` correction loops as evidence of diligence rather than as errors — and rejected traces are routed to a verifier-editor that surgically repairs FDI labels, diagnosis labels, and formatting while preserving the clinical chain of thought; the repaired trace is re-verified before promotion. Both stages are cross-family by construction: the generator and the verifier are routed through separate provider pools spanning different model families (Gemini, MiniMax, NVIDIA NIM, Groq, OpenRouter), so a generator's blind spot is unlikely to be shared by its judge. Because generation is conditioned on privileged information, traces are scanned for *directive leakage* — the demonstrator echoing its hidden instructions in its reasoning — and leaking cases are excised and regenerated under a dual gate (zero-leak regex audit + verifier re-approval); the canonical corpus contains zero leaks.

The resulting corpus (Table 2, Fig. 8) comprises 3,694 verified traces over 1,847 images from two institutions: 1,847 interactive with-tools traces and 1,847 single-turn no-tools traces in exact 1:1 parity — the paired design that lets the evaluation isolate the contribution of the tools (Section 4.1). The with-tools corpus carries 5,144 finding-level ground-truth annotations across cohorts (4,513

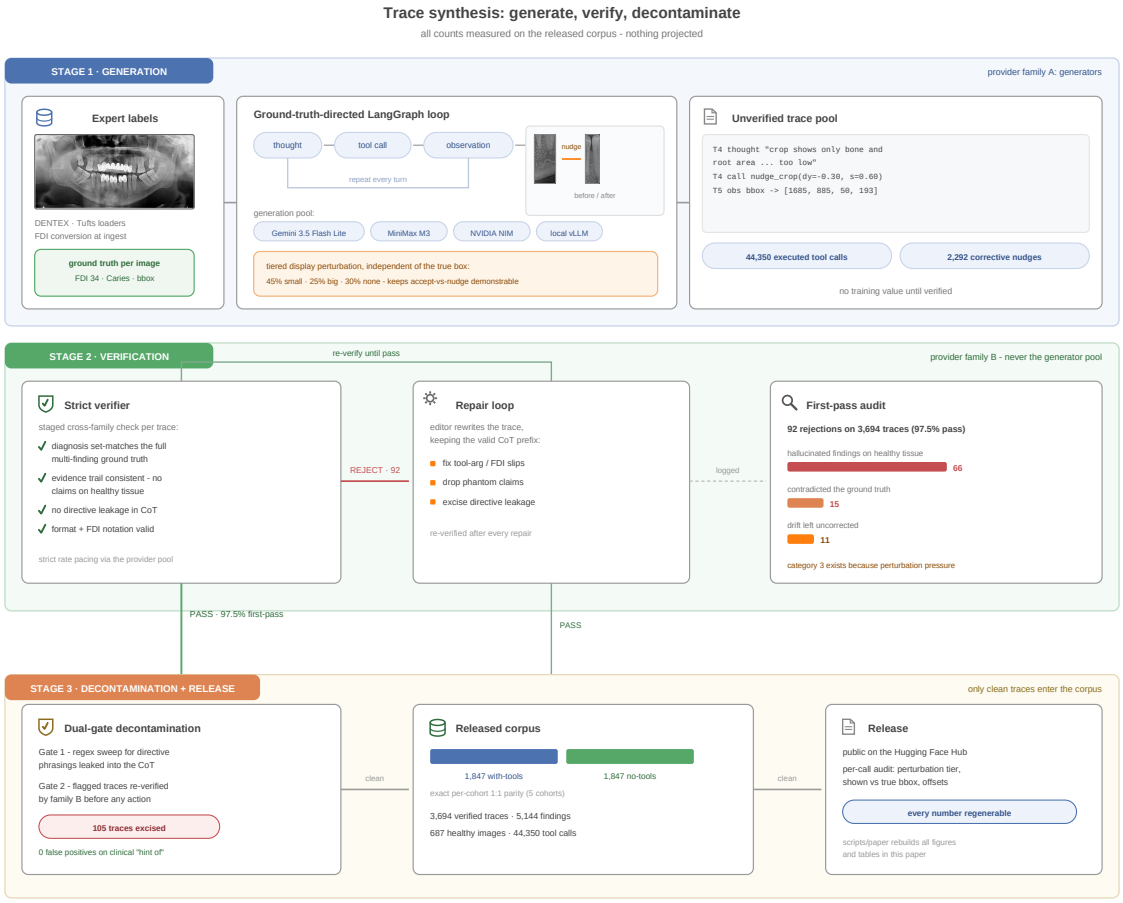


Figure 7: Multi-stage trace synthesis. Generation (provider family A) runs a ground-truth-directed LangGraph loop with real tool execution and the tiered perturbation hook; staged verification (deliberately a *different* provider family) combines a strict clinical pass with a repair-and-re-verify loop; a dual-gate leak audit and corpus assembly produce the frozen canonical release; the audit bars are measured counts. The verifier's 92 first-pass rejections span three categories — the third (uncorrected spatial drift) exists only because the perturbation mechanism creates genuine judgment pressure.

unique: the 202-image Tufts overlap cohort is contained in the 280-image Tufts all-diseases cohort, so its 631 findings are counted twice), spanning caries (2,189), deep caries (578), impacted teeth (604), periapical lesions (789 unique), and three further Tufts lesion classes (353); multi-finding images carry up to 23 annotated findings, and the teacher surveys every finding on the image. Healthy scans are included as negative controls with empty finding sets. The verifier rejected 92 traces across three failure modes: 66 hallucinated findings on healthy scans, 15 ground-truth contradictions (wrong diagnosis class or swapped quadrants), and 11 uncorrected spatial drifts — cases where the teacher diagnosed from a visibly displaced box without issuing a corrective `nudge_crop`. The third category is direct evidence that the perturbation mechanism creates real, auditable judgment pressure rather than a scripted step. Across the with-tools corpus the teacher averaged 12–13 turns and 18–30 real tool calls per trace (Table 2); the full call distribution is shown in Fig. 9.

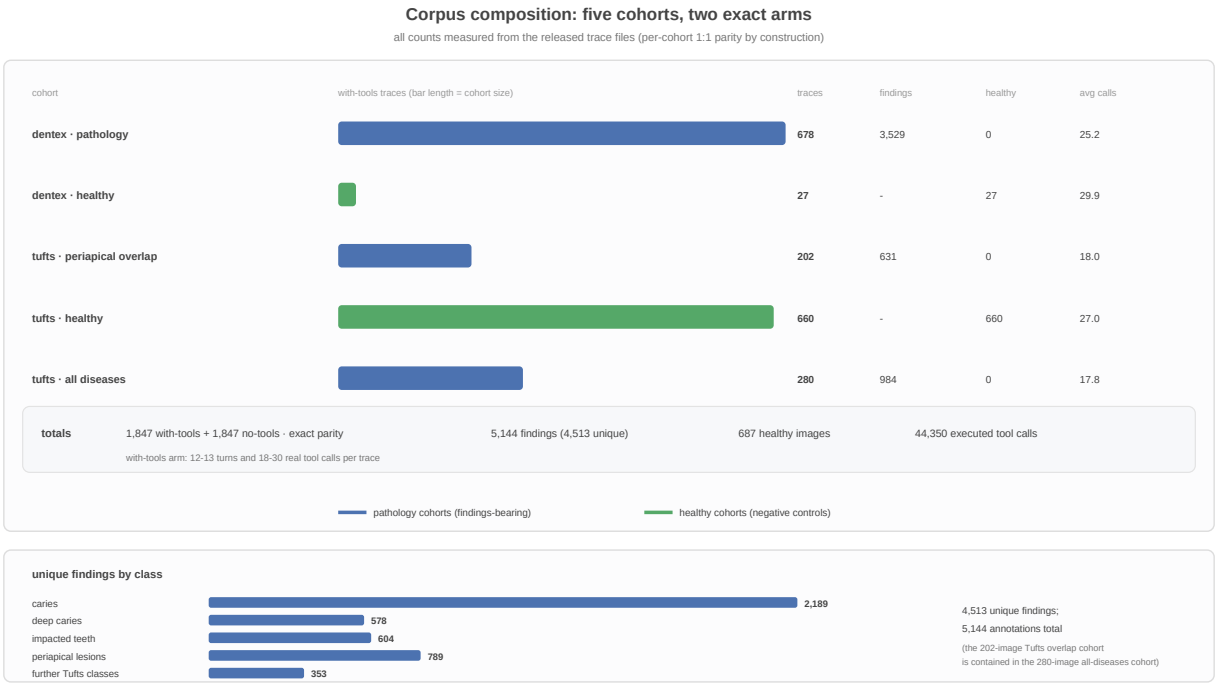


Figure 8: Corpus composition (measured from the released trace files). Five cohorts — three findings-bearing, two healthy negative controls — exist in two exact arms (with-tools and no-tools) with per-cohort 1:1 parity; the with-tools corpus carries 5,144 finding-level annotations (4,513 unique: the 202-image Tufts overlap cohort is contained in the 280-image Tufts all-diseases cohort, so its findings are counted in both). Class mix of the unique findings below; the teacher surveys every finding on an image, and healthy scans carry empty finding sets.

Table 2: The verified corpus, regenerated from the released trace files. With-tools and no-tools cohorts are in exact 1:1 parity (checkmarks). Turns and tool calls are for the with-tools side; findings are counted per cohort (the Tufts overlap cohort is a subset of the Tufts all-diseases cohort, so its findings are counted in both).

Cohort	With-tools	No-tools	Findings	Healthy	Turns	Tool calls
DENTEX pathology	678	678 ✓	3529	0	12.9	25.2
DENTEX healthy (negative controls)	27	27 ✓	0	27	13.4	29.9
Tufts overlap pathology	202	202 ✓	631	0	12.0	18.0
Tufts healthy (negative controls)	660	660 ✓	0	660	12.8	27.0
Tufts 4-disease pathology	280	280 ✓	984	0	11.7	17.8
Total	1847	1847 ✓	5144	687	–	–

3.7 Anatomy of a trace: with tools vs. without

Fig. 10 shows both halves of the paired corpus for one DENTEX image (a carious FDI tooth 34, a single finding). The no-tools trace is one turn: a fluent, well-structured direct diagnosis that happens to reach the right answer, with no evidence trail — the reasoning cannot be checked against anything the model was actually shown, and the model never had the opportunity to be wrong and recover. The with-tools trace for the same image runs 23 turns and 22 real tool calls and demonstrates the full behavioral repertoire the perturbation mechanism is designed to elicit: the grounding box for tooth 34 was inspected, judged wrong ("the crop shows only bone/root area . . . the box appears

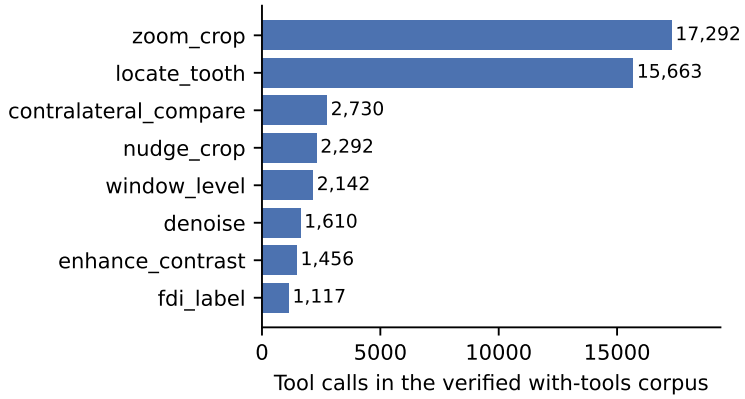


Figure 9: Recorded tool-call attempts over the verified with-tools corpus (44,350 across 1,847 traces, of which 48 were invalid — e.g. a hallucinated `localize_tooth` — and returned a standard error observation to the agent). `zoom_crop` and `locate_tooth` dominate; `nudge_crop` appears 2,292 times as genuine mid-episode correction.

to be too low"), and corrected three times with box-relative `nudge_crop` shifts until re-inspection confirmed the crown; a contralateral symmetry comparison and a contrast-enhanced re-inspection followed before the structured final answer, which matches the expert label.

The paired design is what makes this contrast usable rather than anecdotal. With 1,847 with-tools and 1,847 no-tools traces in exact per-cohort parity, the evaluation can hold the image and the label fixed and vary only the tool interface, isolating what interactive tool use adds over single-pass reasoning (Section 4.1). The no-tools traces are retained deliberately: they teach output format and clinical vocabulary and provide the baseline arm of every comparison — but they are documented as ground-truth-directed single-pass generation (closer to hindsight rationalization than blind discovery), a tradeoff Section 3.6 made explicit rather than hidden.

3.8 Training: SFT then GRPO with a correction-friendly reward

Stage 1 (SFT) fine-tunes Qwen3.5-9B (Bai et al., 2025) with QLoRA (Dettmers et al., 2023) on the verified with-tools traces, teaching the *shape* of the behavior: well-formed reasoning, well-formed tool calls with continuous parameters, and structured multi-finding final answers. Multi-turn trajectories interleave the real crops the tools produced, and loss is masked to assistant tokens only.

Stage 2 (GRPO (DeepSeek-AI, 2025)) optimizes the composite reward

$$R = w_{\text{acc}} R_{\text{acc}} + w_{\text{fmt}} R_{\text{fmt}} + w_{\text{tool}} R_{\text{tool}} + w_{\text{eff}} R_{\text{eff}},$$

with three properties that matter for this task:

Ground-truth-anchored, multi-finding accuracy. R_{acc} is a hard, non-LLM-judge match against expert labels. Predicted findings are matched to annotated findings one-to-one by greedy highest-score-first assignment; each matched pair scores $0.25 \cdot \mathbf{1}[\text{quadrant}] + 0.25 \cdot \mathbf{1}[\text{position}] + 0.50 \cdot \mathbf{1}[\text{diagnosis}]$, and the image-level score is the F1-style harmonic mean of the matched recall and precision — missed findings and hallucinated extra findings both score zero, so spamming guesses cannot free-ride. With one finding on each side this reduces exactly to the single-pair score. This design follows Fan et al. (2026)'s documented reward-hacking failure under an LLM-judge accuracy term: our accuracy signal is not a judge's opinion and cannot be collapsed by calling one tool and stopping.

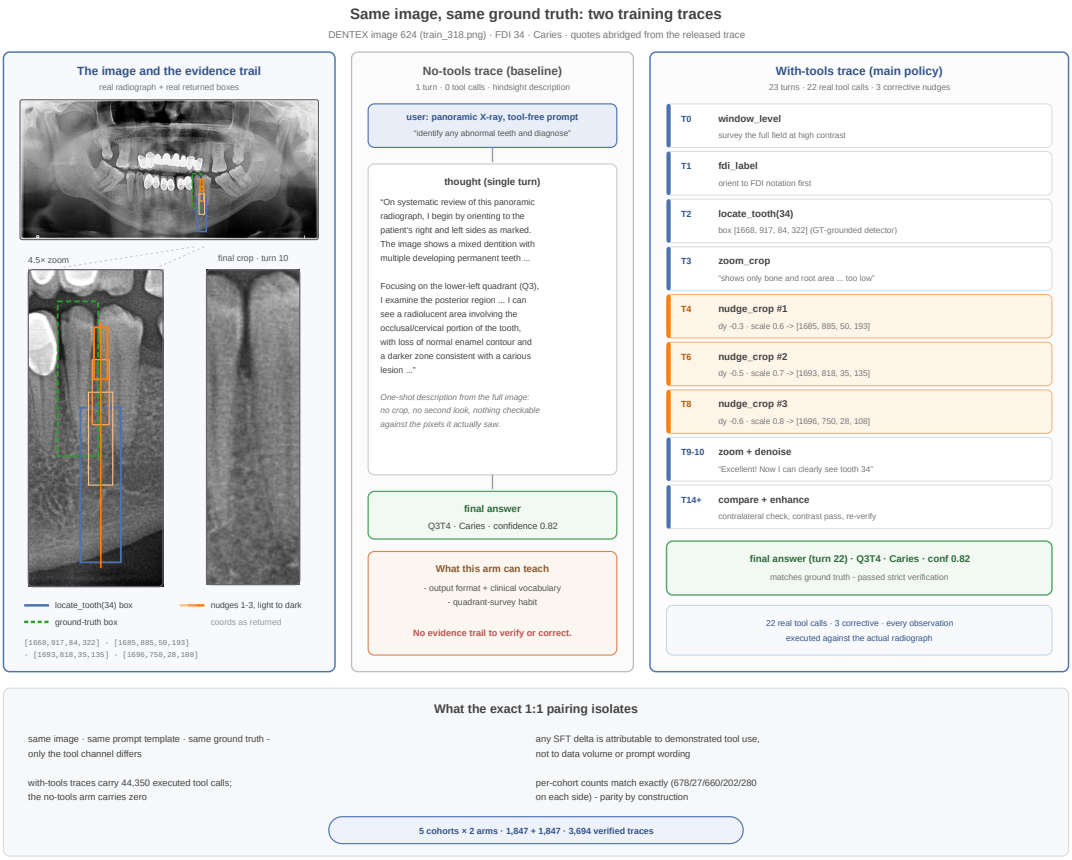


Figure 10: Two training traces for the same image (DENTEX, carious FDI 34) and the same expert label. Left: the real radiograph with the with-tools evidence trail — the returned grounding box (blue), the three corrective `nudge_crop` boxes (orange, light to dark, magnified in the inset; the arrow marks the upward walk from the returned box to the corrected one), and the expert box (green dashed); all coordinates are the values actually returned by the tools, and the right panel shows the crop the teacher finally saw. Middle: the single-turn no-tools trace — fluent, correct, and unverifiable. Right: the with-tools trace (23 turns, 22 real tool calls), in which the grounding box is inspected, judged wrong, corrected three times, and re-inspected before the answer; quotes are abridged from the released traces. The two arms exist in exact per-cohort 1:1 parity, so the evaluation can vary only the tool interface.

Complexity-scaled efficiency that does not fight correction. A naive per-call penalty prices every tool call equally — and therefore prices a legitimate corrective `nudge_crop` the same as aimless spam, directly fighting the behavior Section 3.5 exists to teach. R_{eff} instead derives a reference budget from the trajectory itself (6 calls per distinct located tooth), exempts `locate_tooth` and `nudge_crop` entirely from per-call cost, and charges only budget overruns and exact back-to-back repeat calls (which yield no new information), with a smooth drop-off rather than a hard cliff.

Dual-adapter RL. The frozen SFT adapter serves as the KL reference policy on the same base model, computing the KL penalty by adapter swapping rather than a second model in memory.

Table 3: Zero-shot frontier VLMs on the DENTEX validation set ($N = 50$ panoramic radiographs; set-level matching over all annotated findings). Exact match requires the right tooth *and* diagnosis. ECE = expected calibration error.

Model	N	FDI acc.	Exact acc.	Exact F1	95% CI	ECE
Moonshot Kimi-k3	50	64.0%	**46.0%**	**0.222**	[32.0, 60.0]	**0.214**
Meta Llama 3.2 90B	46	45.7%	37.0%	0.098	[23.9, 52.2]	0.395
Gemini 3.5 Flash	50	62.0%	36.0%	0.188	[22.0, 50.0]	0.560
MiniMax M3	50	62.0%	32.0%	0.110	[20.0, 44.0]	0.493
Gemini 3.5 Flash Lite	50	54.0%	32.0%	0.166	[18.0, 46.0]	0.596
Qwen3.5-9B (our backbone)	49	53.1%	30.6%	0.112	[18.4, 42.9]	0.600
Majority-class floor	50	38.0%	8.0%	0.038	[2.0, 16.0]	–
Meta Llama 3.2 11B	50	26.0%	4.0%	0.010	[0.0, 10.0]	0.678

4 Experiments

4.1 Evaluation protocol

Data and splits. We evaluate on DENTEX's official 50-image validation set (46 pathological images with 182 annotated target teeth, plus 4 healthy controls) as the test set, used exactly once for final numbers. All configuration choices — checkpoint selection, ablation conditions, group-size sweeps — are made on a fixed dev split of ∼70 training images drawn once with a fixed seed, so that no choice leaks into the test result. Healthy images are evaluated with ground truth $\emptyset$: an invented finding on a healthy image is charged to precision, which is where it belongs. Cross-dataset context comes from the Tufts Dental Database (Tufts University School of Dental Medicine, 2023), which contributes training traces and grounding training data but is not used for test-time evaluation in this draft.

Metrics. Findings are matched as a *set*: predictions and annotations are paired one-to-one by optimal assignment (Kuhn, 1955), and a finding is correct at two levels (exact: right tooth and diagnosis; tooth-only: right tooth). Counts are pooled over all findings before computing precision/recall/F1, so a healthy image's invented finding costs precision and images with many findings are not diluted by averaging per-image scores. Uncertainty is quantified by resampling whole *images* (not findings) with replacement; comparisons between conditions are paired, using the same resamples on the images the conditions share. Alongside accuracy we report the fraction of healthy images on which findings were invented, the fraction of episodes ending without a parseable answer, and expected calibration error (ECE) (Guo et al., 2017) between stated confidence and correctness.

Conditions. Every condition is a checkpoint (base, SFT, GRPO) with or without tool access, evaluated under the input geometry used in training (original images for native-trained checkpoints; the canonical views of Fig. 4 for canonically trained ones; tools always operate on the native image) and additionally at original size. The no-tools arms use the *same* SFT and GRPO recipe on the 1,847 single-turn traces, so tool contribution is not confounded with training recipe.

4.2 Zero-shot baselines: the case for looking again

Table 3 establishes the problem the agent architecture addresses. Even the best frontier model misses more than half of the exact findings; the best-calibrated model still carries ECE 0.21 while most

Table 4: Main results on the DENTEX validation set (pooled finding-level, set-matched); "Best zero-shot frontier" is the best row of Table 3. Cells marked [] are pending the first hardware training run; the experimental design and metric definitions are final and frozen (Section 4.1).

Condition	Tools	Exact F1 ↑	Tooth-only F1 ↑	Invented↓
Best zero-shot frontier	–	0.222	–	–
Zero-shot Qwen3.5-9B	–	0.112	–	–
Majority-class floor	–	0.038	–	–
SFT only (no tools)	–	[pending]	[pending]	[pending]
SFT + tools	✓	[pending]	[pending]	[pending]
SFT + GRPO (no tools)	–	[pending]	[pending]	[pending]
VLM-DENTAL (SFT + GRPO)	✓	[pending]	[pending]	[pending]
Supervised specialist (HierarchicalDet), DENTEX: AP 0.372 / AP_{50} 0.601 / AR 0.653 (Hamamci et al., 2024)				

models report 0.85–0.95 confidence on hallucinated positions; and a dense 11B generalist falls *below the majority-class floor*, a statistical collapse rather than a graceful degradation. The failure modes are structural, not stochastic: quadrant inversions from the OPG's left/right display convention, a resolution bottleneck that reduces an early caries to a sub-patch blur, tissue-masking of periapical radiolucencies, and severe overconfidence. Each of these maps onto a specific tool in Table 1 — which is the design rationale of the suite, not a post hoc story: grounding corrects spatial disorientation, `zoom_crop` bypasses the resolution bottleneck, `window_level` and `enhance_contrast` counter tissue masking, and `contralateral_compare` tests symmetry hypotheses before commitment.

4.3 Main results

Table 4 is the paper's primary claim table: does the agentic recipe beat zero-shot inference and the same backbone without tools (H1: tools help; H2: RL helps), and where does it stand against a supervised specialist (H3, framed as a fair test rather than a guaranteed win). The training runs are in progress; all result cells are filled from the frozen evaluation protocol above as runs complete.

4.4 Mechanism analysis: does tiered perturbation elicit self-correction?

Before any training, the corpus itself tests the central design claim of Section 3.5: did the perturbation hook make the *teacher* — an expert-level frontier model with no verbal hints — actually inspect, judge, and correct grounding output, rather than scriptedly nudge? We answer this by re-walking every with-tools trace (8,404 analysed `locate_tooth` returns) and reconstructing, for each located tooth, the box the teacher was shown, the reference box, and every subsequent `nudge_crop` applied to that box through the trace. The shown box is recovered from the retained observation payloads (exact post-perturbation boxes; validated at zero mean coordinate error against reference boxes on the clean tier) or, where context truncation removed the observation, from the teacher's own downstream `bbox` arguments, attributed by nearest center. Each attributed nudge is then classified by direction: *corrective* if its shift moved the working box toward the reference center (cosine > 0.3), *away* if against it, *rescaling* otherwise.

Table 5 and Fig. 11 show the intended gradient, cleanly and monotonically: corrective nudges on 38.4% of large offsets, 13.6% of small ones, and 1.1% of clean boxes. The clean-tier control matters: the teacher does not nudge reflexively — with a correct box it proceeds 98.9% of the time, so the gradient is driven by the *observed* displacement, not a scripted habit. The small tier behaves exactly as designed as a judgment call: nudges there are frequent (23.1%) but only sometimes corrective (13.6%), consistent with the zoom margin often absorbing a small offset. On corrected large offsets,

Table 5: Correction behavior by perturbation tier, over the ground-truth-anchored subset of locate returns (reference box = expert box; pathology cohorts). P(nudge) counts any attributed `nudge_crop`; P(corrective) counts locates with at least one corrective-direction nudge; IoU after is over corrected locates. The clean tier is a control: with a correct box, corrective nudges are rare.

Tier	n	P(nudge)	P(corrective)	IoU before	IoU after	ΔIoU
none	2,338	19.4%	1.1%	1.00	0.27 ± 0.18	-0.73
small	1,337	23.1%	13.6%	0.45	0.45 ± 0.18	-0.01
big	1,493	47.2%	38.4%	0.10	0.24 ± 0.18	+0.15

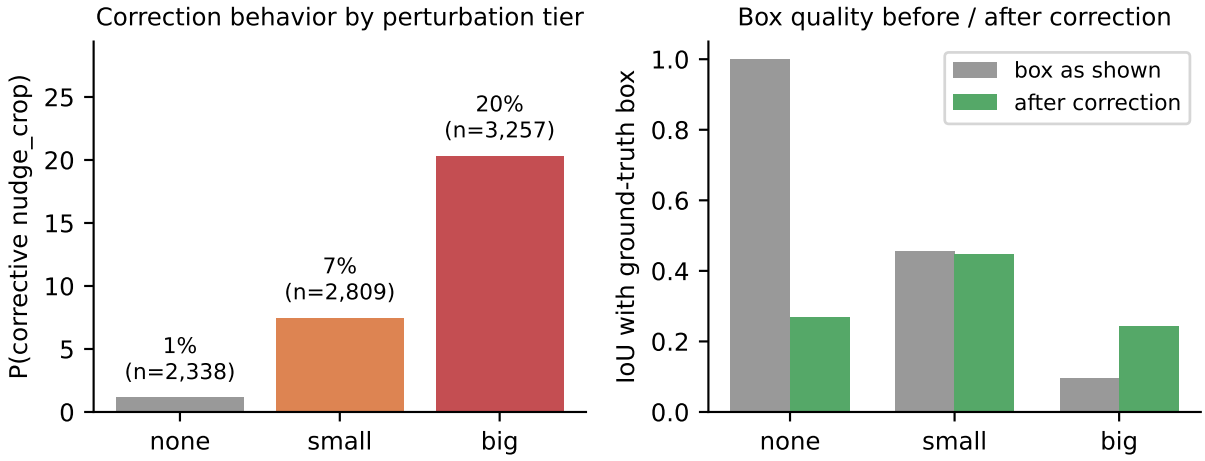


Figure 11: Left: corrective `nudge_crop` usage rises monotonically with perturbation severity (overall corpus, all cohorts). Right: mean IoU with the reference box before and after correction, on the ground-truth-anchored subset. The "none" after-bar reflects the few corrective nudges on clean boxes (1.1% of that tier), which are mostly reframing rescales rather than corrections.

mean IoU against the expert box improves from 0.095 to 0.243 per correction episode — partial on average, because a single nudge rarely lands perfectly — with individual recoveries as large as IoU 0.066 → 0.938 on a DENTEX training image (Fig. 12). The residual gap is not silently absorbed: uncorrected spatial drifts are exactly what the verifier's third rejection category (11 traces) caught and repaired during corpus construction.

Two further observations sharpen the picture. First, the split by reference type: on *detector-anchored* locates (exploratory calls during healthy-scan full-arch surveys, where the reference is the detector's own box), correction is much rarer (5.0% on large offsets vs. 38.4% ground-truth-anchored) — during rapid systematic surveys the teacher surveys broadly and does not inspect each exploratory box as carefully. We report this honestly; the demonstrated skill this paper targets is the ground-truth-anchored pathology workflow. Second, nudge *rescales* skew small (scale < 1): the dominant corrective pattern is tighten-after-shift, matching radiologist behavior of zooming into a suspected lesion after centering it.

4.5 Ablations and robustness checks

All ablation machinery is implemented and runs on a trained checkpoint with nothing retrained; results are reported here as the runs complete.

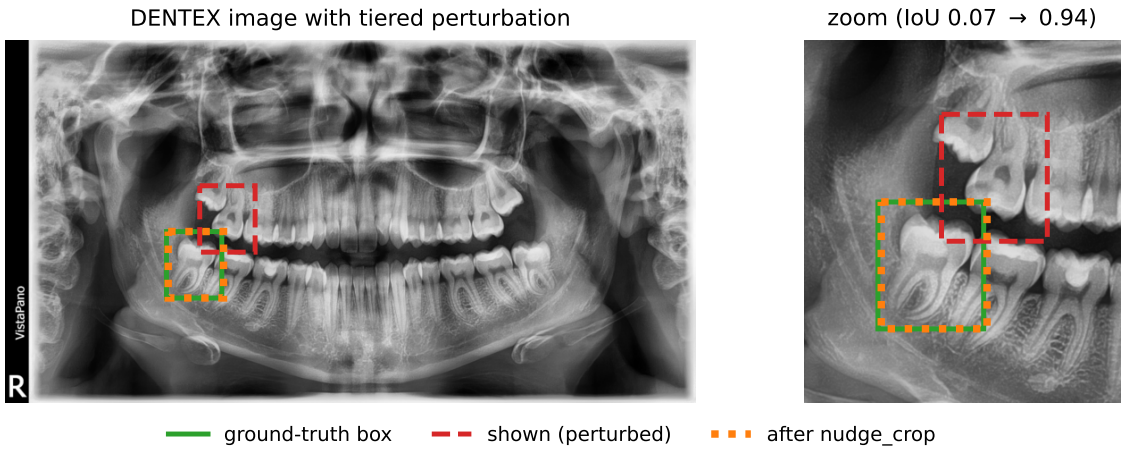


Figure 12: A large-offset perturbation drawn from the training corpus (DENTEX image `train_263`, ground-truth-anchored). The teacher was shown the red box, one tooth away from the expert box (green); it issued `nudge_crop` with a box-relative shift and re-inspected, recovering IoU 0.066 → 0.938 (orange, dotted). This is the demonstrated behavior the student policy is trained to imitate and then reinforce.

Tool-removal probe. A tool is removed from the registry (absent from the prompt's tool list; calls to it return the standard "not recognized" error), and we record both the score drop and how often the trained agent attempts the missing tool. This measures dependence on each tool, including `nudge_crop`; because the checkpoint was trained with the tools present, it is a dependence probe, not the score of a tool-free model — the no-tools training arms in Table 4 serve the latter role. **[pending]**

Grounding-perturbation probe. Every `locate_tooth` box at evaluation time is displaced by a fixed pseudo-random fraction of its own size (the same wrong box for the same image and tooth across conditions, so comparisons stay paired), and we report the IoU of the agent's working box against the annotated tooth before and after its last `nudge_crop`, plus how much of the annotated tooth the last `zoom_crop` actually shows. This is a stress test of the trained policy's self-correction — the deployed version of Section 4.4. **[pending]**

Group-size sweep. GRPO group sizes $K \in \{1, 2, 4, 8, 16\}$ on the dev split, reporting pooled F1 per K with paired comparisons and training cost; reports, never selects. **[pending]**

5 Limitations

No prospective clinician study. Our validity anchor is DENTEX's own dentist-verified ground truth (each image annotated by a final-year dental student and verified by an expert dentist with over 15 years of experience) plus automated reasoning-grounding checks by a cross-family LLM judge. This validates against expert labels, not against a prospective, blinded review by practicing clinicians on new cases. We state this plainly and consider a live reader study the natural next step for a team with clinical access.

Training results are pending in this draft. The training-side evaluations (SFT, GRPO, ablations on the trained agent) are designed, frozen, and implemented, but their runs are not complete at the time of writing; those cells are marked as such rather than extrapolated. The

completed evidence — zero-shot baselines, grounding-tool validation, and the corpus-level mechanism analysis — stands independently, but the paper's end-to-end claims await those runs.

Synthetic perturbations are not real detector errors. The tiered perturbation draws offsets from fixed configured ranges rather than from the grounding model's actual error distribution. This is what makes demonstrations stable as the detector improves (Section 3.5), but it means the training distribution of box errors is synthetic: uniform random offsets with random signs, not the structured errors (e.g., adjacent-tooth confusions, root-tip truncations) a real detector exhibits. The evaluation-time grounding-perturbation probe (Section 4.5) measures robustness to a related synthetic distribution; robustness to the detector's true error modes is best assessed post hoc by swapping in improved detector weights, which our decoupling makes safe for the trace corpus but does not substitute for measuring.

Demonstrated vs. learned correction. Section 4.4 measures the *teacher's* behavior in the demonstrations — the skill the student is trained to acquire — not the trained policy's behavior at inference. The policy-side version of the same analysis is precisely the grounding-perturbation probe, pending the trained checkpoint. Demonstrated behavior is a necessary but not sufficient condition for learned behavior.

Scale and scope. The corpus is 3,694 traces over two institutions and four DENTEX diagnosis classes (plus Tufts' four-class taxonomy); it is anchored to precedents showing a few hundred verified examples suffice for medical RL fine-tuning (Hu et al., 2025), but it is small by data-centric standards, and DENTEX's test set is 50 images — confidence intervals on it are wide, and we report them rather than over-reading point estimates. The tool suite covers radiologist-like inspection primitives; measurement and calibration tools (lesion sizing, periodontal probing analogues) and anomaly detection beyond the four classes are out of scope for this paper.

Automated verification is not expert review. The verifier is a frontier model from a different family than the generator, audited against expert labels — and its rejection taxonomy shows it catches real failures — but automated verification has blind spots that human clinical review would not. We mitigate with cross-family verification, ground-truth anchoring of every correctness claim, and release of the full corpus and audit trail so that the community can re-audit it.

6 Conclusion

We presented VLM-DENTAL, an agentic VLM for panoramic dental radiograph diagnosis built on a principle we argue is the right contract between a reasoning policy and its tools in any domain with learned components: *grounding is a hypothesis to interrogate, not an answer to trust.* Concretely, the policy learns to correct its grounding tool (`nudge_crop`, box-relative and data-only), and it can only learn this because the training signal deliberately exposes tool error: a tiered perturbation mechanism that is decoupled from the detector's real accuracy — so demonstrations never go stale — and that never announces errors verbally — so the learned behavior survives to inference time. The same philosophy appears in the reward: complexity-scaled efficiency that exempts exactly the locate-and-correct calls the task genuinely needs, and a ground-truth-anchored, multi-finding accuracy term that an LLM judge cannot be hacked into.

The completed evidence supports each design decision: zero-shot frontier VLMs fail on DENTEX for structural reasons that map one-to-one onto the tool suite; the grounding tool is strong enough to

be useful and imperfect enough to need correcting; and the perturbation mechanism elicits a clean, monotone correction gradient in 3,694 verified demonstrations (38.4% of large offsets corrected, 1.1% of clean boxes) with recoveries up to IoU 0.066 $\rightarrow$ 0.938 — demonstrated judgment, not scripted steps. The end-to-end training results on the frozen evaluation protocol are the immediate next step; the pipeline, corpus, audit tooling, and protocol are released to make every number in this paper regenerable.

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A Perturbation configuration

The exact configuration used for trace generation is given in Table 6.

Table 6: Exact tiered-perturbation configuration used for trace generation. Offsets are drawn uniformly per axis with a random sign and applied with `nudge_crop`'s own shift-and-clamp math, so every shown box is one the correction tool could produce or fix. The unperturbed box is logged internally per call (`true_bbox`, `perturb_tier`) and never shown to the model.

Tier	Probability	Offset magnitude (per axis, box-relative)	Intent
none	0.45	0	Clean accept-and-proceed demonstration
small	0.25	$U(0.12, 0.28)$, random sign	Genuine judgment call; not resolved by the prompt
big	0.30	$U(0.45, 0.75)$, random sign	Visually self-evident without any verbal cue

Rates are fixed constants, deliberately decoupled from the grounding detector's measured accuracy so that improving the detector never invalidates existing demonstrations (Section 3.5).

B Mechanism analysis: full split by reference type

Table 7 reports the correction-behavior analysis split by reference type; Section 4.4 discusses the ground-truth-anchored rows in the main text.

Table 7: Corrective-nudge behavior split by reference type, regenerated from the released corpus. *GT-anchored*: the locate call was ground-truth-grounded (annotated tooth; pathology cohorts) and the reference box is the expert box. *Detector-anchored*: exploratory calls (dominated by healthy-scan full-arch surveys), where the reference is the detector's own box. During rapid surveys the teacher corrects far less often per box — a real behavioral difference we report rather than average away.

Reference	Tier	n	P(nudge)	P(corrective)	IoU before → after
GT	none	2,338	19.4%	1.1%	1.00 → –
GT	small	1,337	23.1%	13.6%	0.46 → 0.45
GT	big	1,493	47.2%	38.4%	0.10 → 0.24
Detector	small	1,472	2.9%	1.8%	0.37 → 0.38
Detector	big	1,764	6.9%	5.0%	0.09 → 0.15

Recovery rates of the shown box (from retained observations or downstream `bbox` arguments) were 0.97–1.00 in every cell; validation of observation-derived boxes against reference boxes on the clean tier gave zero mean coordinate error.

C Grounding tool validation

Table 8 summarizes the grounding tool's validation.

Table 8: YOLOv8m grounding tool (32 FDI classes), 5-fold cross-validation on the combined DENTEX+Tufts pool (2,339 images, 46,808 boxes) and evaluation on the held-out target benchmark (50 images, 182 annotated targets, greedy one-to-one target matching). "Target mAP50" restricts evaluation to detections on annotated target teeth.

Model	Target mAP50	Target mAP50-95	Precision	Recall@0.50
DENTEX-only (5-fold mean)	0.909	0.638	0.939	0.818
DENTEX+Tufts (5-fold mean)	0.926	0.644	0.963	0.775
Best fold (DENTEX+Tufts, fold 1)	**0.959**	**0.650**	**0.986**	0.797

The precision/recall profile is the intended operating point for a grounding *tool* inside an agent loop: high precision so crops are plausible, non-perfect recall so that correction (Section 3.4) and re-location remain live skills rather than dead code paths.

D Verifier rejection taxonomy

During first-pass verification the independent cross-family verifier rejected 92 of 3,694 traces (97.5% first-pass pass rate), all subsequently repaired and re-verified to a 100% final yield:

1. **False-positive pathology on healthy scans** — 66 rejections (71.7%). The generator hallucinated findings on disease-free scans; repaired to systematic negative surveys committing to `[]`.

2. **Ground-truth contradictions and FDI errors** — 15 rejections (16.3%). Mismatched pathology classes or swapped quadrants; repaired via targeted re-prompting.
3. **Uncorrected spatial drift** — 11 rejections (12.0%). The teacher diagnosed from a visibly displaced grounding box without a corrective `nudge_crop`; repaired with spatial realignment. This category exists *because* of the perturbation mechanism (Section 3.5) and is direct audit evidence that it creates genuine, checkable judgment pressure.

In a separate decontamination pass, 105 traces in which the generator echoed its hidden ground-truth directive in its reasoning were excised and regenerated under a dual gate (zero-leak audit + verifier re-approval); the canonical corpus contains zero directive leaks and exact 1:1 parity between the with-tools and no-tools cohorts.

E Reproducibility

Every number in this paper is regenerable from released artifacts: corpus statistics, tool-usage counts, and the mechanism analysis via `scripts/paper/generate_paper_assets.py` over the released trace files (tables in `paper/generated/`, figures in `paper/figures/`); all figures have SVG sources. The evaluation protocol (Section 4.1) is implemented in the repository's evaluation package; verbatim system prompts are extractable programmatically from the source (`dental_agent/paper/prompts.py`). Traces are hosted on the Hugging Face Hub; model checkpoints and the grounding weights are synced to the same Hub organization.